ABHAY SINGH Data Analyst

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PROFESSIONAL SUMMARY

Results-driven Data Science professional with experience in machine learning, data analytics, and statistical modeling. Proficient in Python, SQL, and Power BI with a proven track record of building ML-powered applications and delivering actionable insights from complex datasets.

EDUCATION

10th and 12th, *Divine Public School, Gorakhpur, Uttar Pradesh*

B.Tech in Computer Science, *SR Institute of Management and Technology, Lucknow, Uttar Pradesh*

SKILLS

Programming Languages
- Python Programming
- SQL
- C Language

Data Engineering & Tools
- Data Cleaning
- Data Preprocessing
- Data Validation
- ETL Pipelines
- Excel (Advanced Functions)
- Statistics

Deep Learning
- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- TensorFlow & Keras

Machine Learning
- Machine Learning
- scikit-learn
- Predictive Modeling
- Statistical Analysis

Database
- MySQL
- Database Design
- Query Optimization

Data Analysis & Visualization
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Exploratory Data Analysis (EDA)
- Data Visualization
- Jupyter Notebook

Web Development
- HTML
- CSS
- JavaScript
- Bootstrap
- Django

EXPERIENCE

Data Science Intern, *Unified Mentor* 08/2024 – 10/2024
Remote
- Analyzed **50,000+ records** using Pandas and NumPy to extract actionable business insights.
- Built and deployed ML models achieving **85%+ accuracy** for classification and predictive analytics tasks.
- Automated data cleaning workflows, reducing manual effort by **40%** and enhancing overall data quality.
- Conducted **Exploratory Data Analysis (EDA)** to identify patterns, outliers, and correlations.

Data Science Internship, *Softpro India* 04/2025 – 06/2026
Lucknow, India
- Engineered ETL pipelines to transform large-scale datasets, improving data processing efficiency by **30%**.
- Developed predictive models using **scikit-learn** to support business intelligence and strategic decision-making.
- Created interactive **Power BI** dashboards, enabling stakeholders to visualize key metrics and trends.
- Implemented data validation frameworks to ensure data integrity across 10+ disparate data sources.

CERTIFICATES

- MS Excel Online Training Course
- Comprehensive Data Visualization Course: Matplotlib in Python
- Learn SQL- Structured Query Language
- Python Data Science Course With Numpy, Pandas and Matplotlib

PROJECT

Django Chatbot Web Application | Python, Django, Scikit-Learn, NLP

- **Chatbot Development**: Developed an chatbot web application using **Python**, **Django**, and **Scikit-Learn** forautomated user interaction and response generation.
- **NLP & Feature Engineering**: Implemented tokenization, stop-word removal, and **TF-IDF vectorization** fortext preprocessing and intent prediction.
- **Machine Learning Models**: Built and optimized **Intent**, **Category**, and **Flag** classification models using **Scikit-Learn pipelines** and **LinearSVC**.
- **Model Integration & Deployment**: Integrated trained **ML** models into the **Django** backend using **Pickle** for real-time chatbot inference and response handling.
- **Frontend & Session Management**: Designed responsive UI using **HTML**, **CSS**, **Bootstrap**, and **Django** Templates with session-based chat management.

Entertainer Career Analysis | Python, Pandas, Matplotlib

- **Multi-Source Data Integration**: Consolidated and cleaned multiple Excel-based datasets (Basic,Breakthrough, and Last Work Info) using **Pandas** to create a unified view of entertainer career lifecycles.
- **Exploratory Data Analysis (EDA)**: Performed comprehensive EDA to identify career breakthrough patterns,performance distributions, and achievement trends using **Python**.
- **Statistical Visualization**: Developed categorical and descriptive visualizations using **Matplotlib and Seaborn** to communicate complex career metrics to non-technical stakeholders.
- **Insight Synthesis**: Authored a technical summary report documenting actionable insights regardingentertainer attributes and longevity.

Car Marketplace & Management System | Python, Django, Machine Learning

- **Full-Stack Web Application Development:** Developed a full-stack car marketplace platform for buying, selling, renting, and servicing vehicles using **Python, Django, HTML, CSS, Bootstrap,** and **JavaScript.**
- **Machine Learning-Based Car Price Prediction:** Built a **Machine Learning**-based car price **prediction system** using **Scikit-Learn** and integrated it into the Django application.
- **AI-Powered Car Recommendation Engine:** Designed an AI-powered car **recommendation engine** using **Deep Learning embeddings** and **Cosine Similarity** for personalized vehicle suggestions.
- **Multi-Role Management System:** Implemented ro**le-based authentication** and dashboards for Customers, Staff, and Admins with order, service, and request management features.
- **Data Processing & Feature Engineering:** Managed **data preprocessing, feature engineering,** and **model integration** using **Pandas, NumPy, Joblib,** and **Hugging Face** model hosting.
- **Production Deployment & DevOps**:Deployed the application on **Render** with **PostgreSQL**, **Gunicorn**, **WhiteNoise**, and **cloud-hosted ML** models.

Netflix Content Analysis & Visualization | Python, Pandas, Matplotlib

- **Engineered Data Pipelines**: Developed a Python-based workflow using **Pandas and NumPy** to clean andpreprocess raw Netflix datasets, resolving missing values and duplicate records for data integrity.
- **Exploratory Data Analysis (EDA)**: Executed comprehensive EDA to identify distribution patterns betweenMovies and TV Shows, uncovering trends in release timelines and regional content production.
- **Visual Insight Generation**: Leveraged **Seaborn** and **Matplotlib** to build visualizations for genre trends andcountry-wise content saturation, summarizing findings into an actionable insights report.

Netflix AI Recommendation System | Python, Django, Scikit-Learn, Pandas, Machine Learning

- **Recommendation System Development**: Developed an AI-powered movie recommendation web application using Python, **Django**, and **Scikit-Learn** for personalized Netflix-style movie suggestions.
- **Deep Learning & Similarity Modeling**: Implemented **Deep Learning Movies Encoder** and **Cosine Similarity** algorithms to generate intelligent content-based movie recommendations.
- **Data Processing & Feature Engineering**: Performed Netflix dataset **cleaning**, **preprocessing**, **text normalization**, and **feature extraction** using **Pandas** and **NumPy**.
- **Model Integration & Deployment**: Integrated serialized **DL** recommendation models into Django backend using **Joblib** with automated model downloading from **Hugging Face** during deployment.
- **Frontend & Database Management**: Designed responsive Netflix-inspired UI using **HTML**, **CSS**, **Bootstrap**, and **JavaScript** with **PostgreSQL/ SQLlite** database integration and recommendation history tracking.
- **Cloud Hosting & Production Deployment**: Deployed the complete **Django** application on **Render** with **PostgreSQL**, **WhiteNoise static** handling, **Gunicorn** server configuration, and **cloud-hosted DL model** support.